

Sensors Laboratory Experiment -2

Temperature Sensor Unit Using NTC Thermistor

Roll No. - ES24f1100064

Name : Arpit Katiyar

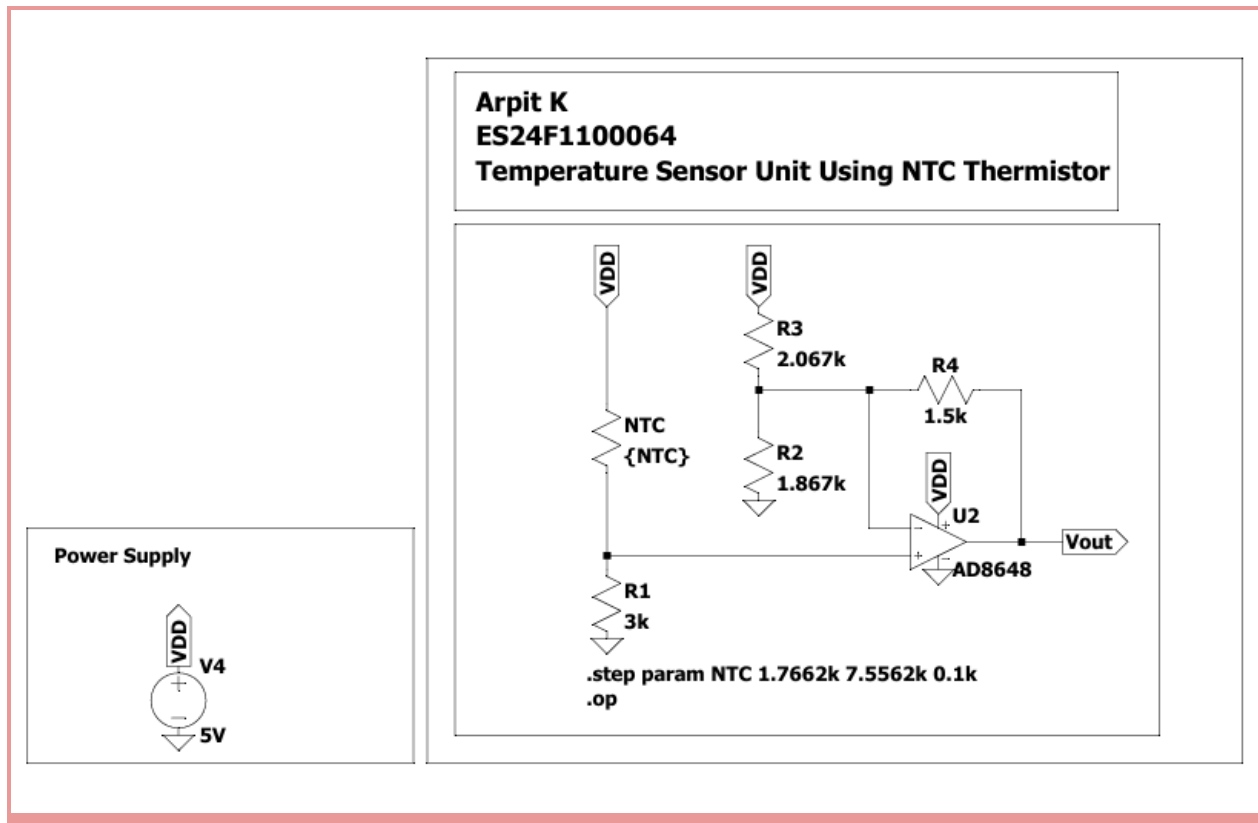
INTRODUCTION

This experiment focuses on designing and testing a temperature sensor unit using an NTC (Negative Temperature Coefficient) thermistor. The thermistor's resistance decreases with increasing temperature, making it useful for temperature-sensing applications. The sensor unit includes a signal conditioning circuit to convert resistance variations into a measurable voltage output. In this experiment, we will design, simulate, and implement the circuit, followed by experimental validation using real temperature measurements.

CIRCUIT DESIGN

The experiment consists of a signal conditioning circuit built using the following components to measure a temperature range of 30°C to 60°C mapped to the voltage range of 0.1 V to 4.5 V.

Schematic of NTC



This temperature sensing circuit uses a resistor in series with a negative temperature coefficient (NTC) thermistor to form a voltage divider, which has the effect of producing an output voltage that is linear over temperature. The circuit uses an op amp in a non-inverting configuration with inverting reference to offset and gain the signal, which helps to utilize the full ADC resolution and increase measurement accuracy.

Circuit Design Calculations :-

The transfer function for the circuit is given by :-

$$\frac{V_{out}}{V_{dd}} = \frac{R_1}{R_{NTC} + R_1} \times \frac{(R_2 \parallel R_3) + R_4}{(R_2 \parallel R_3)} - \frac{R_4}{R_3}$$

Resistance of NTC Thermistor at temperature T (kelvin) is given by :-

$$R_T = R_0 e^{\beta \left(\frac{1}{T} - \frac{1}{T_0} \right)}$$

Where β is the material constant, R_0 is nominal resistance at temperature (kelvin) T_0 From datasheet of the particular NTC.

$$\beta = 4100$$

$$R_0 = 10k \text{ ohm}$$

$$T_0 = 25^\circ\text{C}$$

Therefore,

$$R_T = 10000 \times e^{4100 \left(\frac{1}{T+273.15} - \frac{1}{25+273.15} \right)}$$

For this experiment the values of for temperature range of from 30°C to 60°C with $R_0 T_0$ increment of 5°C are (all the values are in ohms) :-

1. $R_{30} = 7.96 \text{ k}$

2. $R_{35} = 6.39 \text{ k}$

3. $R_{40} = 5.17 \text{ k}$

4. $R_{45} = 4.22 \text{ k}$

5. $R_{50} = 3.46 \text{ k}$

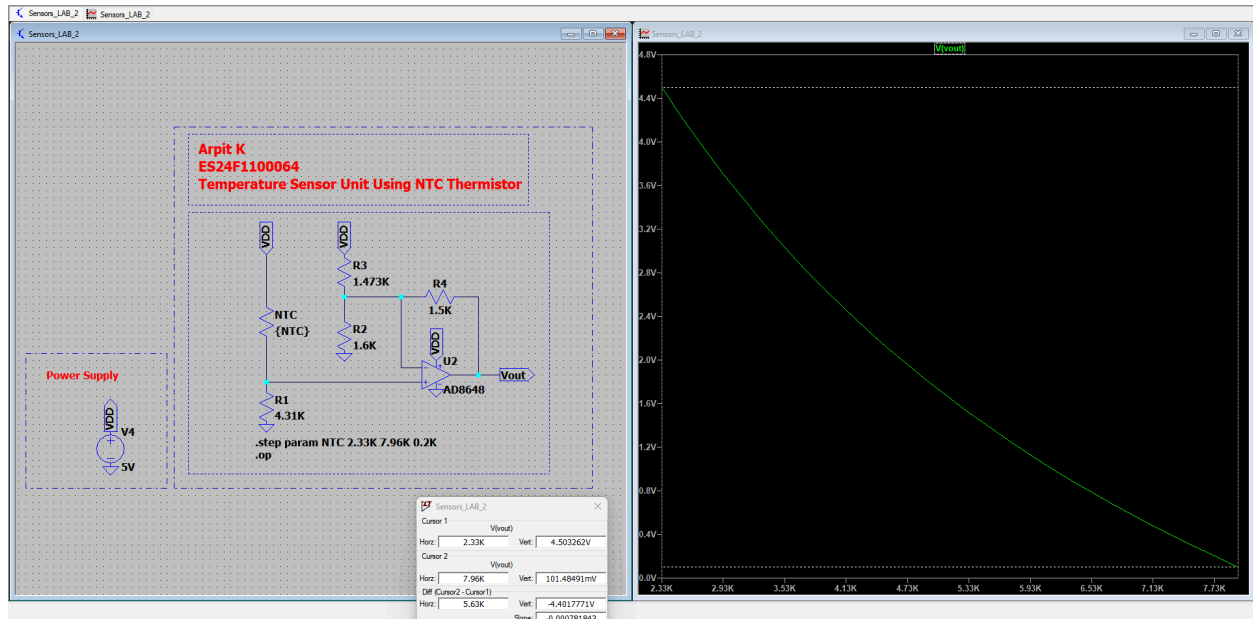
6. $R_{55} = 2.85 \text{ k}$

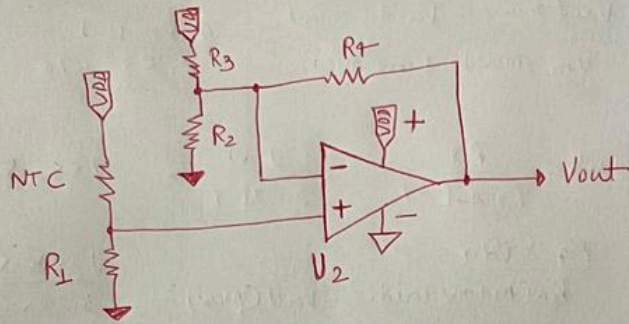
7. $R_{60} = 2.33 \text{ k}$

Let V_T be the voltage at temperature T :

1. $V_{30} = 0.09 \text{ V}$
2. $V_{35} = 0.86 \text{ V}$
3. $V_{40} = 1.63 \text{ V}$
4. $V_{45} = 2.40 \text{ V}$
5. $V_{50} = 3.14 \text{ V}$
6. $V_{55} = 3.84 \text{ V}$
7. $V_{60} = 4.49 \text{ V}$

LT Spice Simulation :-





Voltage Divider .

$$V_{in} = V_{DD} - \frac{R_1}{R_T + R_1} V_{DD}$$

DC Transfer function .

$$\frac{V_{in} - V_{out}}{R_2} + \frac{V_{in} - V_{DD}}{R_3} + \frac{V_{in}}{R_4} = 0$$

Multiply by R_2 $(V_{in} - V_{out}) + \frac{R_2}{R_3}(V_{in} - V_{DD}) + \frac{R_2}{R_4}V_{in} = 0$

$V_{out} = V_{in} \left(1 + \frac{R_2}{R_3} + \frac{R_2}{R_4} \right) - \frac{R_2}{R_3} V_{DD}$

$$A = 1 + R_2 \left(\frac{1}{R_3} + \frac{1}{R_4} \right), \quad B = \frac{R_2}{R_3} V_{DD}$$

$$V_{out} = A \cdot V_{in} - B$$

$$R_{NTC(max)} = 7.97K$$

$$R_{NTC(min)} = 2.33K$$

Now $R_o = \sqrt{R_{NTC(max)} \cdot R_{NTC(min)}} = \sqrt{7.97K \times 2.33K} = 4.33K$

Now $V_{in(min)} = \frac{V_{DD} \times R_1}{R_{NTC(max)} + R_1} = \frac{5 \times 4.33K}{7.97K + 4.33K} = 1.76K$

$V_{in(max)} = \frac{V_{DD} \times R_1}{R_{NTC(min)} + R_1} = \frac{5 \times 4.33K}{2.33K + 4.33K} = 3.23V$

$$G_{ideal} = \frac{V_{out(max)} - V_{out(min)}}{V_{in(max)} - V_{in(min)}} = \frac{4.5 - 0.1}{3.23 - 1.76} = 2.98$$

$$R_2 || R_3_{ideal} = \frac{R_2}{G_{ideal} - 1} = \frac{1.5K}{2.98 - 1} = 755.85 \Omega$$

$$R_3 = \frac{R_2 \times V_{dd}}{V_{in(max)} \times G_{ideal} - V_{out(max)}} = \frac{1.5 \times 3}{3.23 \times 2.98 - 4.5} = 1.45K$$

$$R_2 = \frac{(R_2 || R_3)_{ideal} \times R_3}{R_3 - (R_2 || R_3)_{ideal}} = \frac{755.85 \times 1.45K}{1.45K - 755.85} = 1.576K$$

$$\frac{V_{out}}{V_{dd}} = \left[\frac{R_1}{R_{NTC} + R_1} + \frac{R_2 || R_3 (1 + W_c R_4) + R_4 - \frac{R_4}{R_3}}{R_2 || R_3} \right] \left(\frac{1}{1 + W_c R_4} \right)$$

$$C = \frac{1}{W R_4} = \frac{1}{2\pi f R_4} = \frac{1}{2 \times 3.14 \times 50 \times 1.5K}$$

$$C = 2.122 \mu F$$

If a capacitor is used in parallel to a feedback resistor for noise reduction. If the cutoff frequency is 50 Hz, the noise component at the cutoff will have half power, and will filter higher frequencies as well.

Linearization Calculation :-

To linearize the characteristics, the NTC has to be the average value of the range.
($R_L = 4212.771 \text{ ohm}$)

1. $R_T || R_L$ for 30 deg = 2756 Ω
2. $R_T || R_L$ for 35 deg = 2540 Ω
3. $R_T || R_L$ for 40 deg = 2322 Ω
4. $R_T || R_L$ for 45 deg = 2106 Ω
5. $R_T || R_L$ for 50 deg = 1897 Ω
6. $R_T || R_L$ for 55 deg = 1698 Ω
7. $R_T || R_L$ for 60 deg = 1511 Ω

Calculations for component values (after linearization) :-

$$R_{NTC(max)} = R_{NTC(90^\circ)} = 2.75K; R_{NTC(min)} = R_{NTC(60^\circ)} = 1.51K$$

$$R_1 = \sqrt{R_{NTC(max)} \times R_{NTC(min)}} = \sqrt{2.75K \times 1.51K} = 2.04K$$

$$V_{in(min)} = \frac{V_{dd} \times R_1}{R_{NTC(max)} + R_1} = \frac{5 \times 2.04K}{2.75K + 2.04K} = 2.12V$$

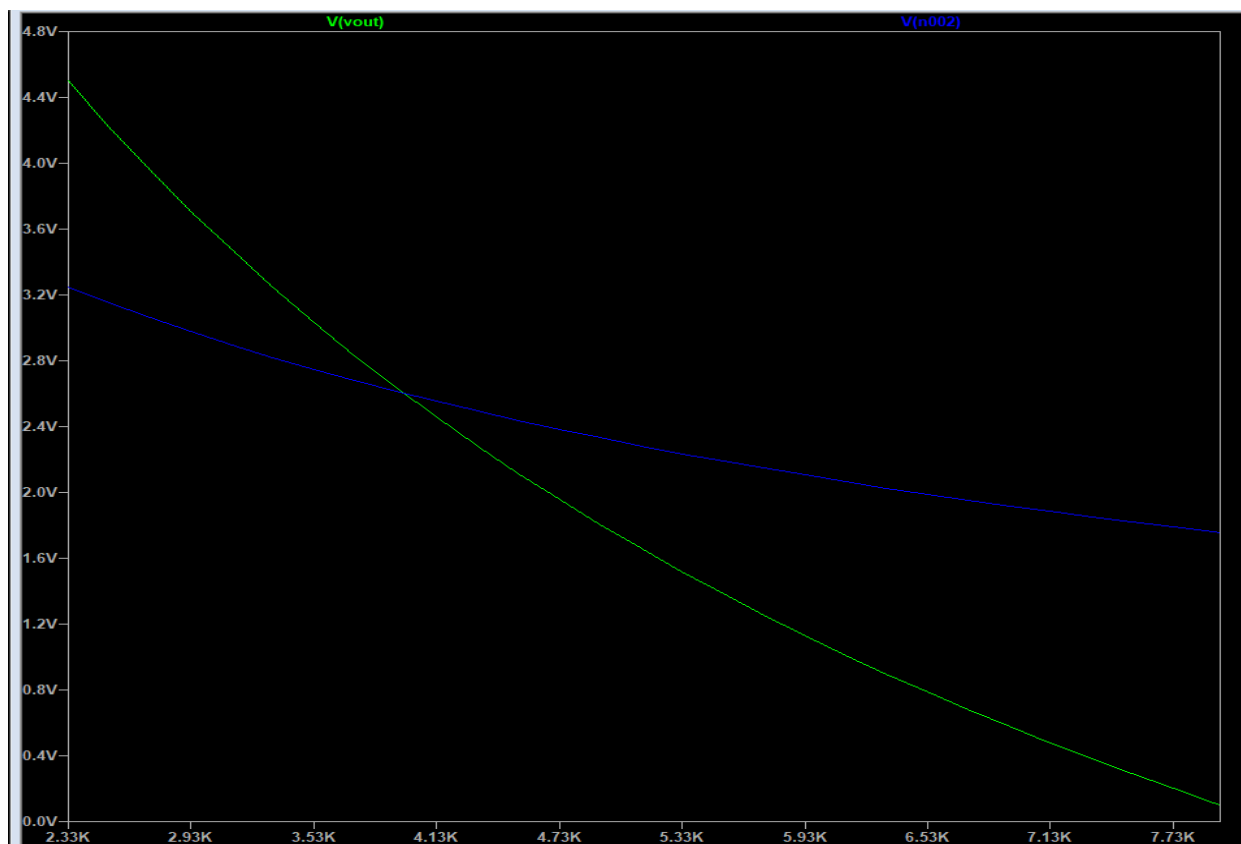
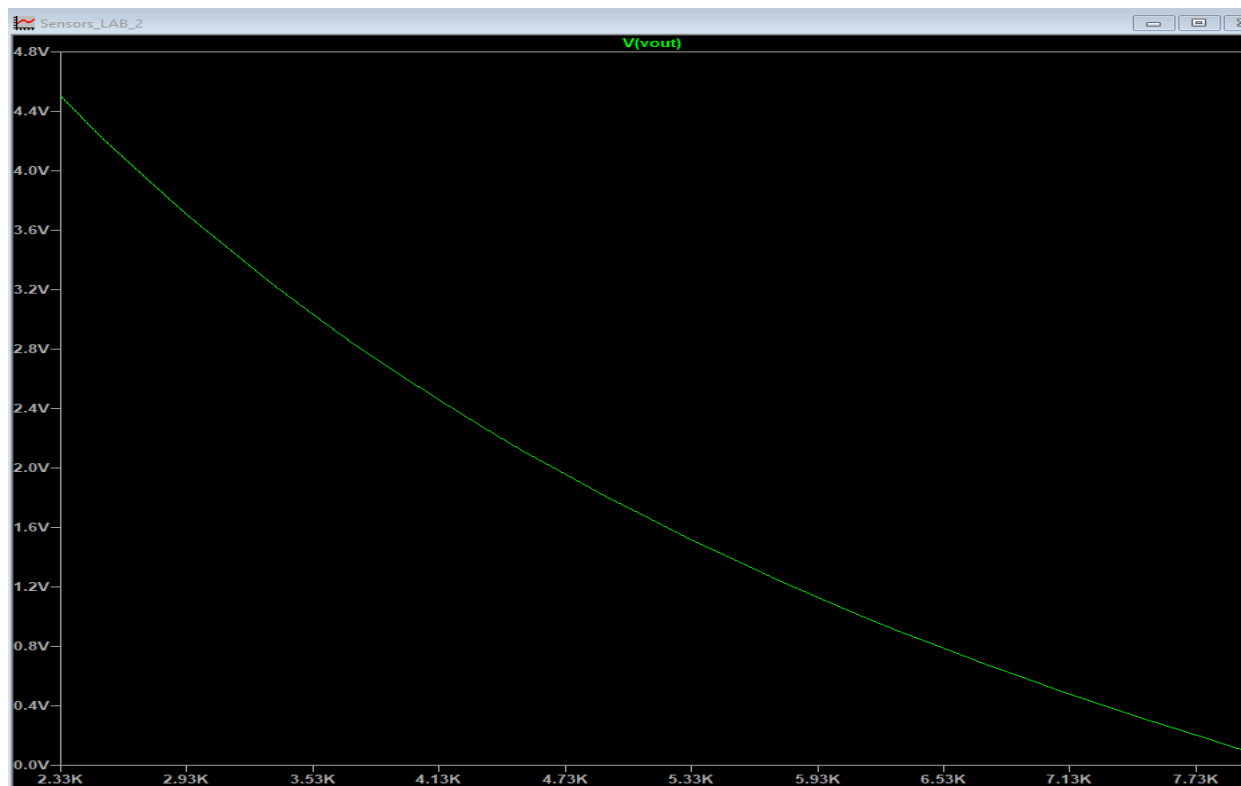
$$G_{ideal} = \frac{V_{out(max)} - V_{out(min)}}{V_{in(max)} - V_{in(min)}} = \frac{4.5 - 0.1}{2.87 - 2.12} = 5.87$$

$$(R_2 \parallel R_3)_{ideal} = \frac{R_4}{G_{ideal}} = \frac{1.0K}{5.87-1} = 308\Omega$$

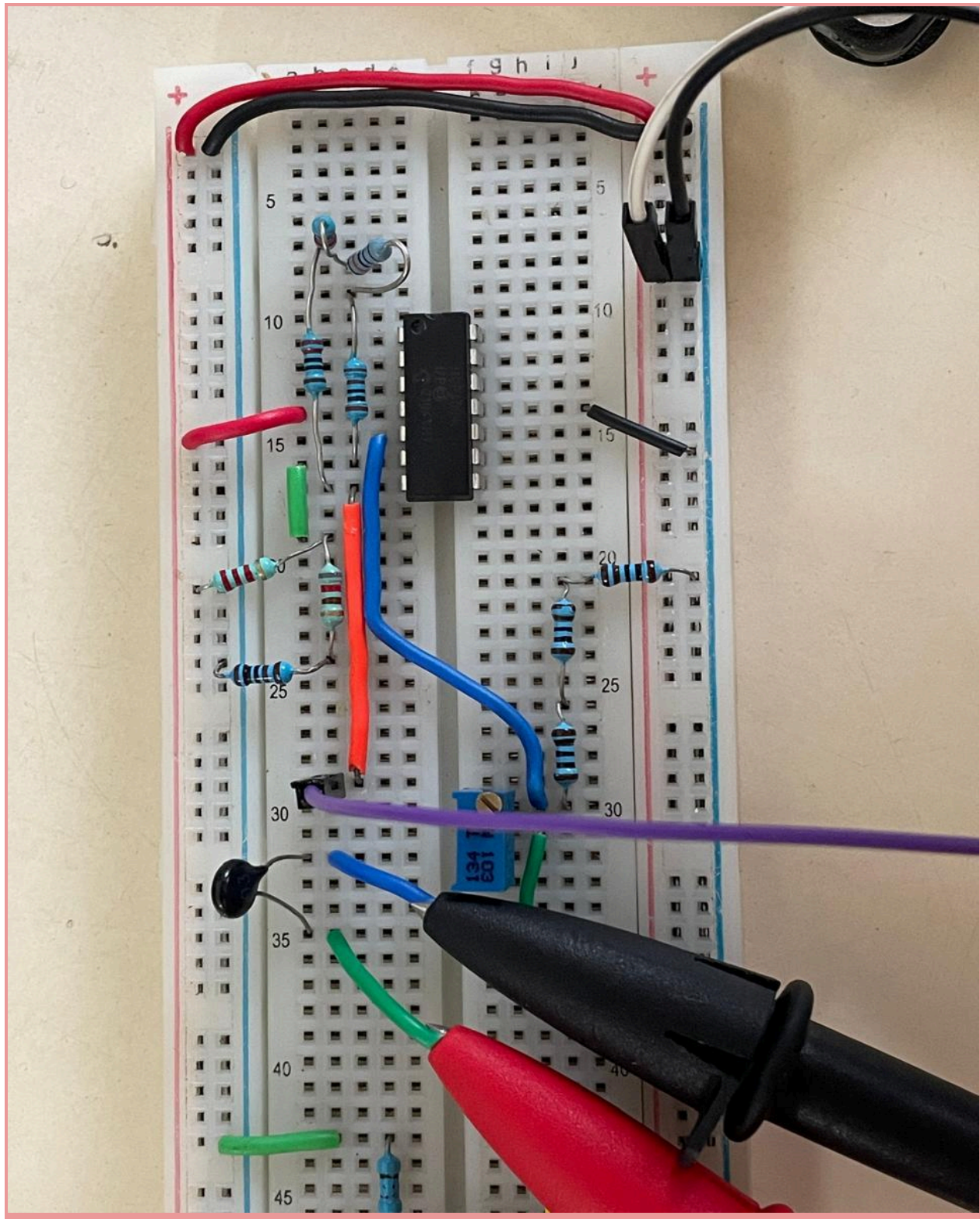
$$R_3 = \frac{R_4 \times V_{dd}}{V_{in(max)} \times G_{ideal} - V_{out}} = \frac{1.5K \times 5}{2.87 \times 5.87 - 4.5} = 607.57\Omega$$

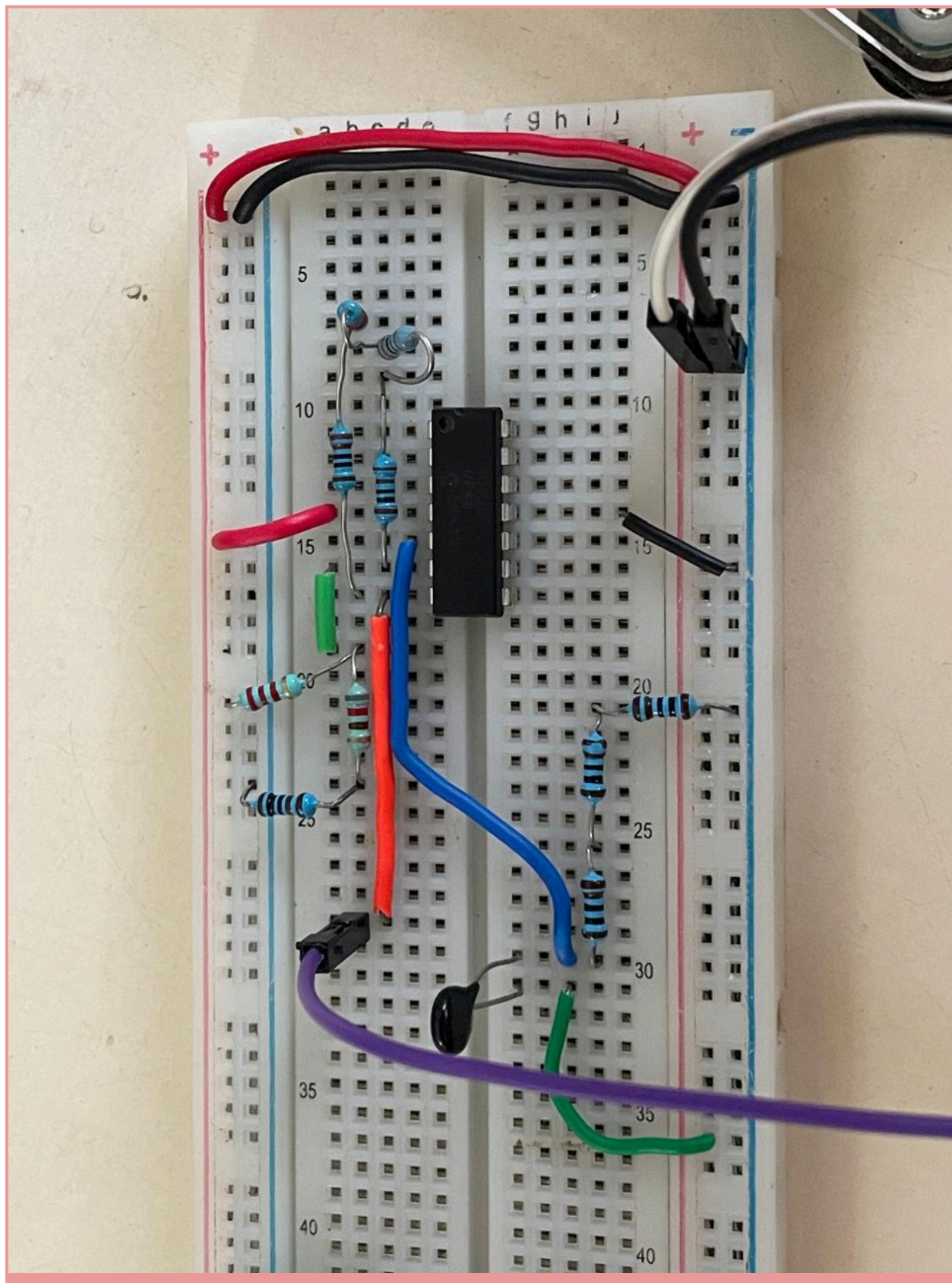
$$R_2 = \frac{(R_2 \parallel R_3)_{ideal} + R_3}{R_3 (R_2 \parallel R_3)_{ideal}} = \frac{308 \times 607.57}{607.57 - 308} = 624.66\Omega$$

V – R characteristic Plot and values of V for different R :

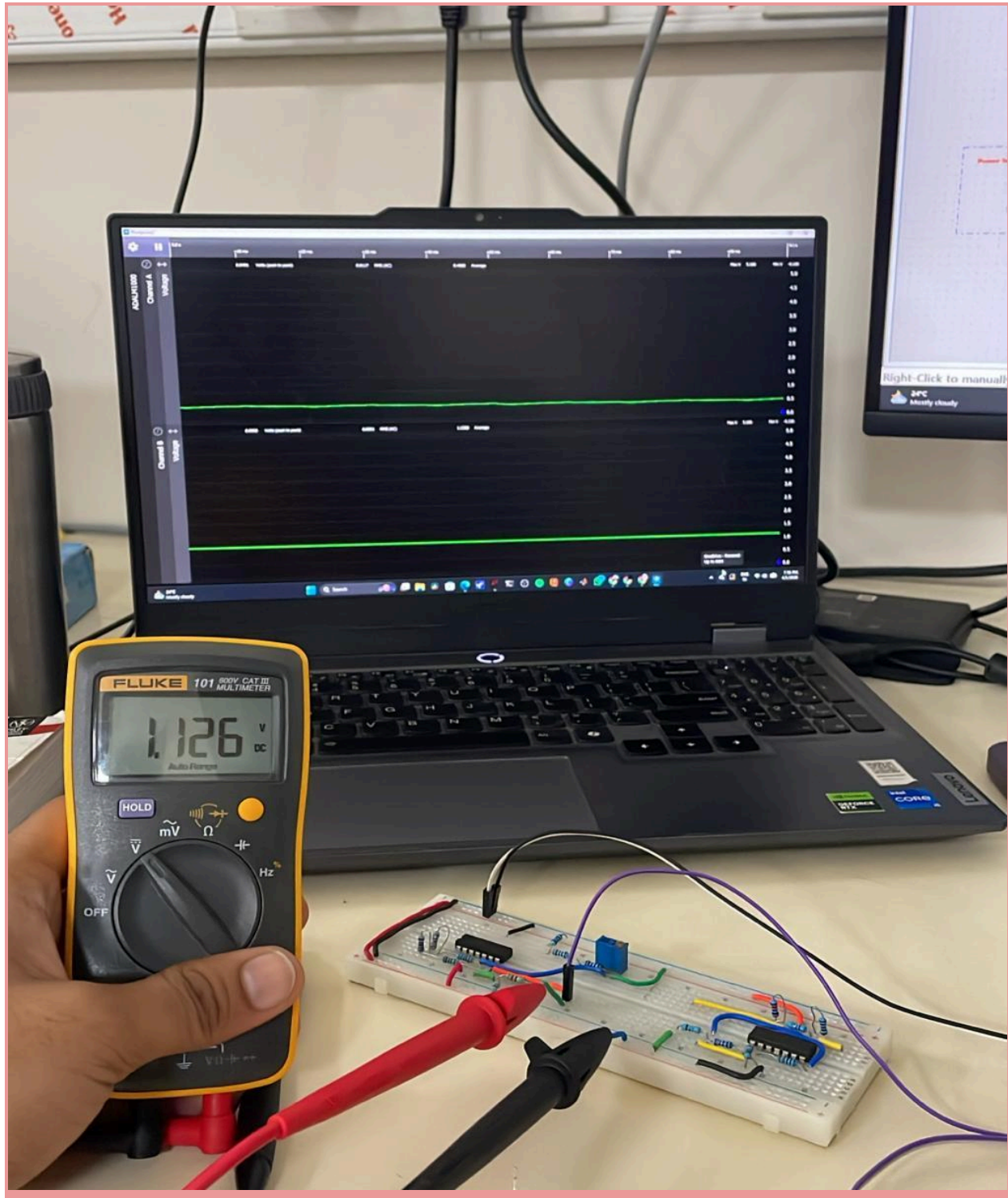


BreadBoard Circuit :





$\text{Temp} = 40$, $R_{\text{ntc}} = 5.16$, $V_{\text{out}} = 1.126$



Room Temperature Monitor	
Time	Temperature
12:00	31
12:30	30.5
01:00	31.2
01:30	32
02:00	32.5
02:30	33
03:00	33.4
03:30	33.6
04:00	32.5
04:30	32
05:00	31.8

Conclusion

This experiment successfully demonstrated the design and testing of an NTC thermistor-based temperature sensing unit. The circuit effectively converted the thermistor's resistance change with temperature into a measurable voltage output, validated through simulation and experimental room temperature monitoring with thermometer cross-checking. While a linear resistance-temperature relationship was observed within the tested range, further work on non-linearity and calibration could enhance accuracy for broader applications. This experiment provided valuable practical experience in sensor interfacing and analog signal processing for temperature measurement.